

From 2D to 3D: Facilitating Single-Finger Mid-Air Typing on Virtual Keyboards with Probabilistic Touch Modeling

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ABSTRACT

Mid-air text entry on virtual keyboards suffers from the lack of tactile feedback, bringing challenges to both tap detection and input prediction. In this poster, we demonstrated the feasibility of efficient single-finger typing in mid-air through probabilistic touch modeling. We first collected users' typing data on different sizes of virtual keyboards. Based on analyzing the data, we derived an input prediction algorithm that incorporated probabilistic touch detection and elastic probabilistic decoding. In the evaluation study where the participants performed real text entry tasks with this technique, they reached a pick-up single-finger typing speed of 24.0 WPM with 2.8% word-level error rate.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction techniques—Text Input; Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual Reality

1 INTRODUCTION

Text entry on head-mounted displays (HMDs) is key to support general access to various VR/AR applications. However, commercial systems are prone to input errors. Users need to input carefully, significantly limiting the input performance and naturalness. Input prediction algorithm [1] plays an essential role in modern text entry techniques, which has been proved effective in numerous scenarios. There are also works that tried to incorporate such algorithms into mid-air typing to facilitate the input performance. However, the requirement of considerable learning effort or typing experience limited the applicability of these techniques in real use.

In this poster, we demonstrated the feasibility of single-finger typing on a virtual keyboard in VR, which requires minimal prior knowledge of the user and introduces little learning effort. Therefore, we believe this technique has a great potential for efficient and natural text input on commercial HMD products. Aiming at the critical challenge of the absence of tactile feedback during mid-air typing, we first collected 12 users' single-finger typing data on different sizes of keyboards, which shed light on their typing behavior patterns. Based on that, we proposed an input prediction algorithm that took the uncertainty in tap detection into the calculation. We also reported the result of an evaluation study in which the participants performed real text entry tasks and reached an input speed of 24.0 WPM. We hope that this work could draw more attention to

probabilistic input prediction and inspire more researchers to further improve the performance of single-finger mid-air typing.

2 EXAMINING THE TYPING BEHAVIOR

We first examined the users' single-finger mid-air typing behavior, as it yielded distinct patterns from that on touchscreens. We recruited 12 right-handed participants (9 male, 3 female) from the campus with an average age of 22.5 (SD = 1.32). We used an Oculus Quest 2 as the apparatus. To avoid the effect of extra factors (e.g., location and angle), according to a pilot study with another 8 participants, we placed the virtual keyboard at a fixed position in the virtual scene that was 40cm away from, and 25.2cm lower than the user's head. Furthermore, the keyboard plane was set perpendicular to the line between the helmet and the keyboard centroid. Three levels of keyboard sizes were tested: *small* (8.1cm × 4.0cm), *medium* (12.3cm × 5.8cm), and *large* (16.5cm × 7.5cm).

During the study, each participant completed 3 sessions of text entry tasks in random order, corresponding to three keyboard sizes. Each session contained 3 blocks. In each block, the participants entered 10 phrases randomly sampled from the MacKenzie and Soukoreff phrase set [3]. They were instructed to type with the index finger of their right hands "as fast and as naturally as possible". During typing, the intersection between the keyboard and their index fingers was not required, and no feedback was provided, which ensured a most natural typing experience. However, they were asked to retype the words if they felt they had committed errors.

2.1 Results

We manually labeled the touches from the continuous typing data according to the log data and the recording video. For each character, we removed outliers that were more than three standard deviations away from the collected centroid on the keyboard plane (small: 177 (2.0%), medium: 187 (2.0%) and large: 190 (2.1%)).

The average typing speed [2] was 23.2 WPM (SD = 4.5), 22.6 WPM (SD = 5.2) and 24.3 WPM (SD = 3.7) for increasing keyboard sizes respectively. RM-ANOVA found no significant effect of *keyboard size* on typing speed ($F_{2,22} = 1.27, p = .30$). Meanwhile, 26.2% (SD = 18.2%), 33.7% (SD = 21.0%), and 35.4% (SD = 24.5%) of all taps intersected with the keyboard plane for increasing keyboard sizes, respectively, with no significant difference found among them ($F_{2,22} = 2.47, p = .11$). This low ratio and great standard deviation suggested that simply inferring taps based on the intersection between the finger and the keyboard plane was not applicable.

3 INPUT PREDICTION ALGORITHM

Aiming at the challenges of tap detection and input prediction due to the ambiguous typing behavior during single-finger mid-air typing, we proposed an algorithm that generalized the classical Bayesian decoding algorithm [1] by introducing probabilistic touch detection and elastic probabilistic decoding. Given the continuous 3D location of the tapping finger $I = I_1 I_2 \dots$, where $I_i = [x_i, y_i, z_i]^T$. The goal was to find the word $W = W_1 W_2 \dots W_m$ in a dictionary with the

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highest probability $P(W|I)$, where W_i denoted the i th character of W . For tap detection, we trained an SVM model with RBF kernel, which used only the z coordinate (perpendicular to the keyboard) of I , and predicted the probability of I_i being within a tap (denoted as p_i). We then merged consecutive frames with $p_i > 0.8$ into one tap, and generated a list of candidate taps $T = T_1 T_2 \dots T_n$, where T_i was the representative frame within the i th tap (i.e., the lowest point). According to Bayesian theory, $P(W|I) = P(W|T) \propto P(T|W) \times P(W)$. We calculated $P(T|W) = D_{n,m}$, and

$$D_{i,j} = \begin{cases} D_{i-1,j-1} \times (1 - P_{ins} - P_{omi}) \times P(T_i|W_j) & \text{-- matching} \\ D_{i-1,j} \times P_{ins} & \text{-- insertion} \\ D_{i,j-1} \times P_{omi} & \text{-- omission} \end{cases} \quad (1)$$

where $D_{0,0} = 1$, P_{ins} and P_{omi} modelled the false positive and false negative rate of the tap detection algorithm, respectively. Further,

$$P(T_i|W_j) = P([x_k, y_k, z_k]|W_j) \times p_k \quad (2)$$

where k was the index of T_i in I . We could infer that the above algorithm took the uncertainty in tap detection into the calculation, and generalized the classical statistical decoding algorithm to handle insertion and omission errors. In implementation, we set P_{ins} and P_{omi} to 0.01 according to the test of the tap detection algorithm on the collected tap data. Moreover, we restricted that the algorithm tolerates at most two insertion or omission errors within each word, to ensure real time performance. We built a unigram language model using the top-10000 frequent words in the ANC [4] data to estimate $P(W)$, which was summarized over 22M words of written American English. When calculating $P([x_k, y_k, z_k]|W_j)$, we used a Bivariate Gaussian distribution for each key based on the collected data, and ignored the z value due to its low signal to noise ratio.

4 PERFORMANCE EVALUATION

We conducted a user study to evaluate the performance of the above algorithm in real text entry tasks. We recruited 12 right-handed participants (9 male, 3 female) with an average age of 22.8 (SD = 0.9). 8 of them have participated in the previous study. The participants completed 5 blocks of text entry tasks using our technique on Oculus Quest 2, with no warm-up session. In each block, they entered 10 phrases randomly sampled from the MacKenzie and Soukoreff phrase set [3]. They were instructed to type "as quickly and as naturally as possible", and could choose to correct the typing errors or leave them uncorrected freely. A two-minute break was enforced between different blocks.

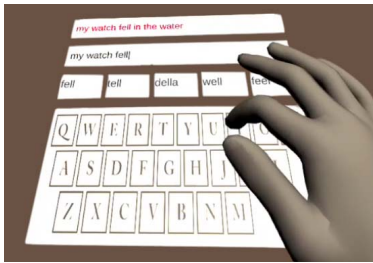


Figure 1: Experiment platform.

During typing, the keyboard showed the top-5 candidate words in the order of probability (see Fig. 1, the top-1 candidate was shown in the input block, and the first candidate showed the raw input). We did not employ space and backspace keys, as hitting them may be confused with hitting character keys. Instead, we designed pinch gestures between the thumb, the index finger, and the middle finger to replace them, respectively. Users could also tap on other candidate words to select them.

4.1 Results

Fig. 2 showed the text entry speed of all the participants. The average speed in different blocks was 24.0 WPM (SD = 3.6), 24.4 WPM (SD = 2.8), 25.4 WPM (SD = 3.6), 27.5 WPM (SD = 3.7) and 29.4 WPM (SD = 3.4) respectively. Significant effect of *block* was found on typing speed ($F_{4,44} = 8.53, p < .0001$). The average speed in block 1 reached 24.0 WPM, showing a satisfying pick-up input performance. The speed in block 5 was 22.5% faster than that in block 1, and has not converged yet, the 5 users with the highest input speed in block 5 even reached an average speed of 32.3 WPM (SD = 1.6), which proved the potential of single-finger mid-air typing.

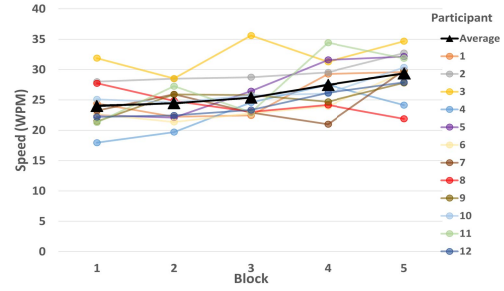


Figure 2: Text entry speed of all participants.

The average uncorrected word-level error rate in each block was 2.8% (SD = 3.4%), 3.7% (SD = 3.9%), 4.3% (SD = 3.2%), 4.3% (SD = 3.3%) and 3.7% (SD = 3.4%), respectively. No significant effect of *block* was found ($F_{4,44} = 0.508, p = .54$). Recalling the results on input speed, we could find that the users could improve their typing speed without sacrificing accuracy, which confirmed the learnability of our technique.

5 CONCLUSION

Unlike typing on touchscreens, we found that the users tended not to intersect with the keyboard plane during mid-air typing. More different typing patterns could also be found given deeper analysis of the collected typing data. Meanwhile, we demonstrated the feasibility of probabilistic touch modeling in this typing scenario. However, a formal investigation of the possible model factors and more advanced techniques (e.g., a better language model) would be helpful for further exploring the limit of such techniques. Our results demonstrated that efficient text entry in mid-air was achievable, and single-finger typing was a potential solution, despite the absence of tactile feedback.

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